

LECTURE - 18

Friday.
3/4/26.

HyperGeometric Distribution: For Lot problems BUT without replacements; means the k changes w each pick.
• This has all 5 conditions of "Binomial distribution" (lec. 16) except that here: the prob. of success (p) changes and hence trials are dependent.

No. of Items = N

One kind (interested) = k

Other kind (not interested) = $N - k$

Sample size (drawn) = n

P.D.F.:

$$P(X=x) = h(x; \underbrace{N, n, k}_{\text{parameters}}) = \frac{{}^k C_x \cdot {}^{N-k} C_{n-x}}{{}^N C_n}$$

Q: A bag of balls contains 9 red balls and 6 Black balls. Sample of 4 balls selected without replacement ^{tells its hypergeo.}. Let X be no. of red balls in the sample drawn, then find prob. distn. of X .

$$N = 9 + 6 = 15, \quad k = 9, \quad n = 4$$

$$X = 0, 1, 2, 3, 4$$

→ we'll choose minimum/low value of x b/w ' k ' and ' n '. like here $n < k$ ($4 < 9$) ∴ x goes till 4.

$$P(X=0) = h(0; 15, 4, 9) = \frac{{}^9 C_0 \cdot {}^{15-9} C_{4-0}}{{}^{15} C_4} = 1/91$$

$$P(X=1) = h(1; 15, 4, 9) = \frac{{}^9 C_1 \cdot {}^{15-9} C_{4-1}}{{}^{15} C_4} = 12/91$$

till $x=4$

answer:

x	0	1	2	3	4
$h(x; 15, 4, 9)$	1/91	12/91	...	...	...

NOTE: If we've to draw 3 red & 1 black balls then we'll be taking K of our choice. Like if $K=9$ (red balls) then 1 black ball will be chosen from $N-K \rightarrow 15-9=6$ so ~~it~~ it doesn't matter.

$$p = \frac{K}{N}, \quad q = 1 - \frac{K}{N}$$

$$\mu (\text{mean}) = np$$

$$\text{variance} = \sigma^2 = np \cdot q \cdot \left(\frac{N-n}{N-1} \right)$$

★ If N becomes very large; Hypergeo. distri. becomes Binomial Distri.

Q: Names of 6 men & 5 women written on slips & placed in bag.

Four names are drawn. What is prob. 2 men & 2 women drawn?

$$N = 6 + 5 = 11, \quad K (\text{interested in men}) = 6 \uparrow, \quad n = 4, \quad x = 2 (\text{exactly 2 men}).$$

$N-K$ (failure $\rightarrow$ women) = 5 $\downarrow$ could've been opposite

$$P(X=2) = h(2; 11, 4, 6) = \frac{{}^6C_2 \cdot {}^5C_2}{{}^{11}C_4} = \boxed{5/11}$$